Homework 1

"JavaScript and the DOM manipulation"

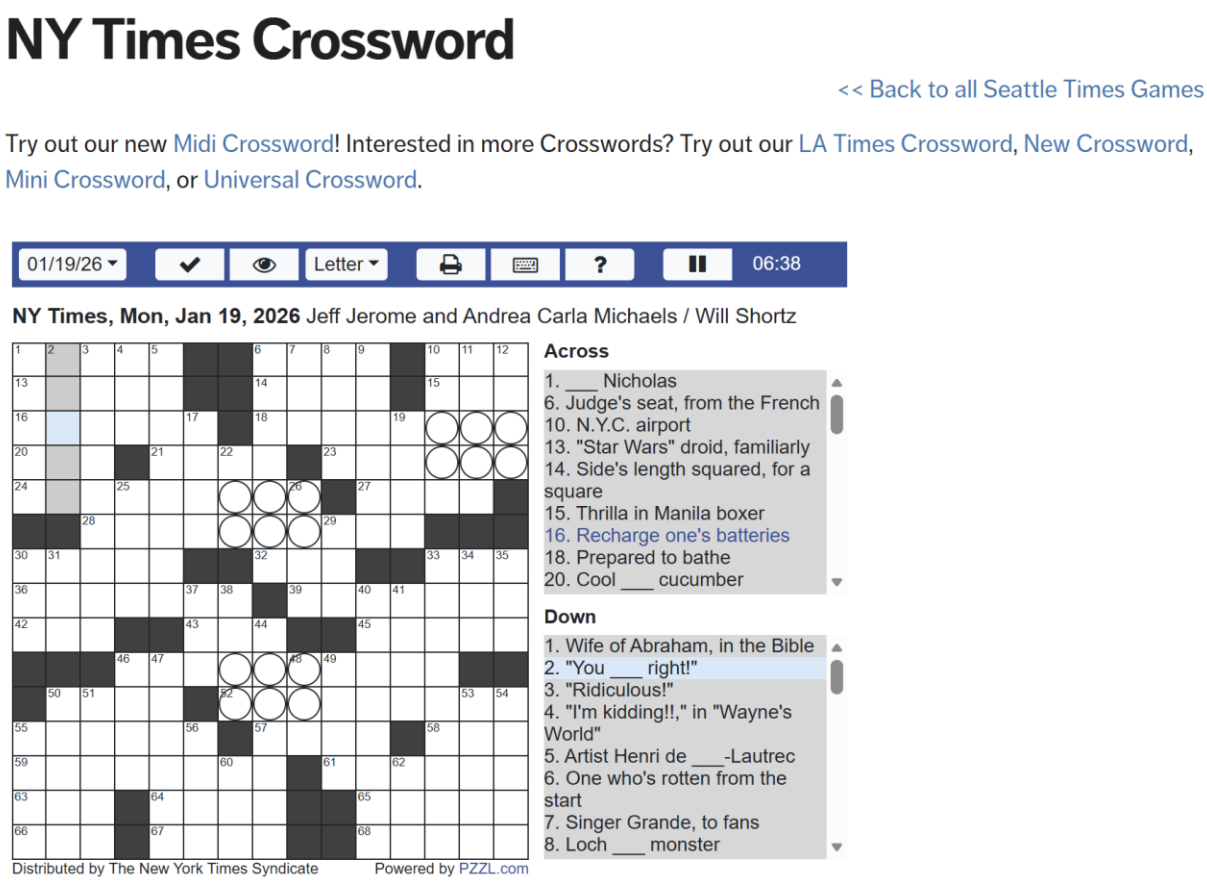
The following problem set is worth 100 points total. Please submit a zip file containing all relevant code to the Canvas drop box by the due date of the assignment.

Requirements

This first problem set will get you familiar with JavaScript as well as the manipulation of the DOM. You will be creating two interactive games using JavaScript.

Crossword Puzzle (40pts)

Each day The New York Times publishes a crossword puzzle in their newspaper. The same puzzle is also available online in interactive form:



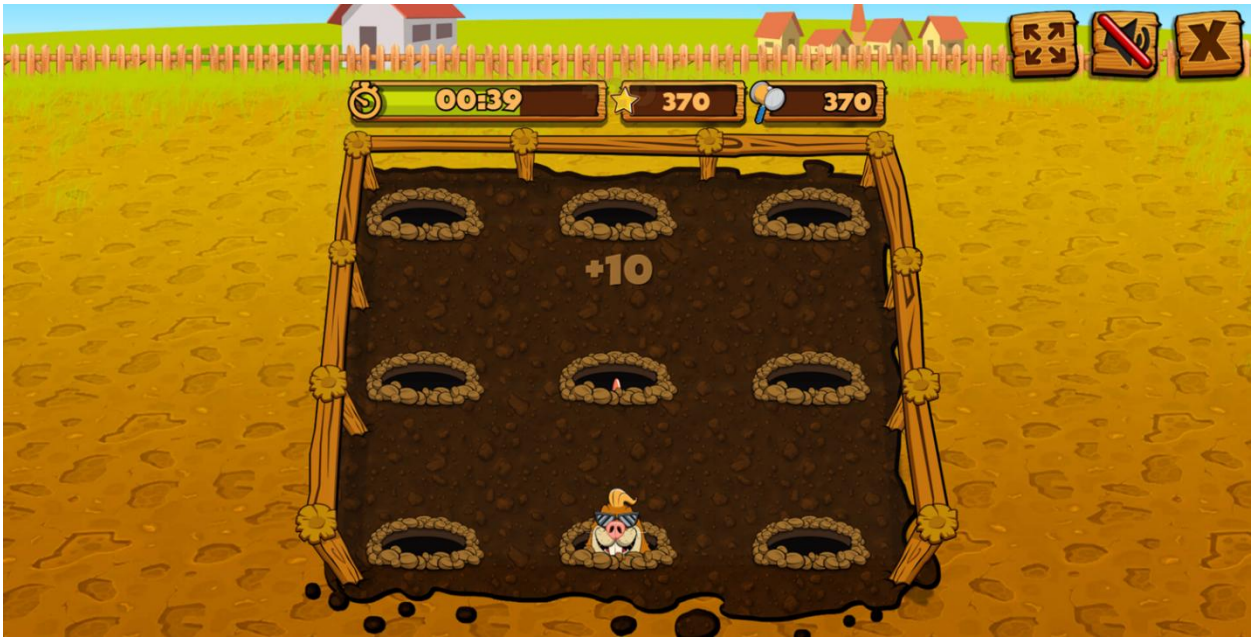
Requirements

For this assignment, you are to randomly produce an interactive crossword puzzle each time the user loads the window. The user should be able to then play the game, clicking on a cell where the user is then able to type in a letter.

- Give the user the option to change difficulty of the game. Difficulty is defined as follows:
    - **Easy**. 10x10 grid with 5 questions across and 5 questions down
    - **Intermediate**. 15x15 grid with 10 questions across and 10 questions down
    - **Hard**. 20x20 grid with 15 questions down and 15 questions across
- If the user correctly completes the puzzle, highlight all text cells green to designate a victory status
- Hardcode the clues (and their answers) in JavaScript data structures behind the scenes. (Research the internet if you need ideas for clues).

Whack-a-Mole (60 pts)

The second part of this assignment is to familiarize yourself with JavaScript's timer functions, notably **setInterval**, **setTimeout** and **cancelTimeout**.



Rules of the game

Whack a mole is a game where the player must correctly "whack" as many moles that come out of their holes in a set period. If the player whacks a mole correctly (e.g. clicks on the graphic), points are added to their score. If the player whacks an incorrect mole (clicks on an empty cell), points are deducted from the score. When the timer is up the game is over and the user sees their score.

Implementation Details

Like the prior problem, you will need to use randomization in your solution.

- Generate a 3x3 game grid initially and when the user decides to start the game, give the player 45 seconds in total
- Randomly have the mole (or moles) appear in differing squares after each second tick
- Give the user three difficulty options (Easy, Medium, Hard) where the parameters behind the mole behavior are adjusted accordingly
- Each mole that appears will be present for a random number of seconds

- If the user correctly clicks on a mole, add +10 to their running score
- If the user incorrectly clicks on a square, deduct -10 from their running score
- When the game is over, display the player's final score
- If you need a mole graphic, one is added to the Canvas assignment

Submission Requirements

For the submission, submit a zip file with all assets needed for me to run your two applications. Feel free to use any technique used in class (either DOM manipulation or jQuery.) Submit using the Canvas drop box before the due date.